

Title: AI-Based Fake Profile Detection in Social Networks

Abstract

The rapid growth of social networking platforms has led to a significant rise in fake, bot-operated, and impersonation profiles that spread misinformation, conduct scams, manipulate engagement metrics, and compromise user trust and platform integrity. Traditional rule-based detection methods rely on a limited set of hand-crafted heuristics and often fail to keep pace with increasingly sophisticated fake account behaviour. This project proposes an AI-Based Fake Profile Detection framework that leverages machine learning and deep learning techniques to automatically identify fake, bot, or impersonated profiles on social networks with high accuracy and reliability.

The proposed system analyse multiple categories of profile signals, including profile metadata (username patterns, profile picture authenticity, bio content, account age), behavioural patterns (posting frequency, activity timing, engagement ratios), and network-level graph features (follower-following ratios, community structure, connection density). Text-based attributes such as bio and post content are processed using NLP techniques and transformer-based embeddings (e.g., BERT), while structural attributes are modelled using graph-based methods such as Graph Neural Networks (GNNs). Classical machine learning models (Random Forest, SVM, Logistic Regression) and deep learning models (CNN, LSTM) are trained and benchmarked to identify the most effective detection approach.

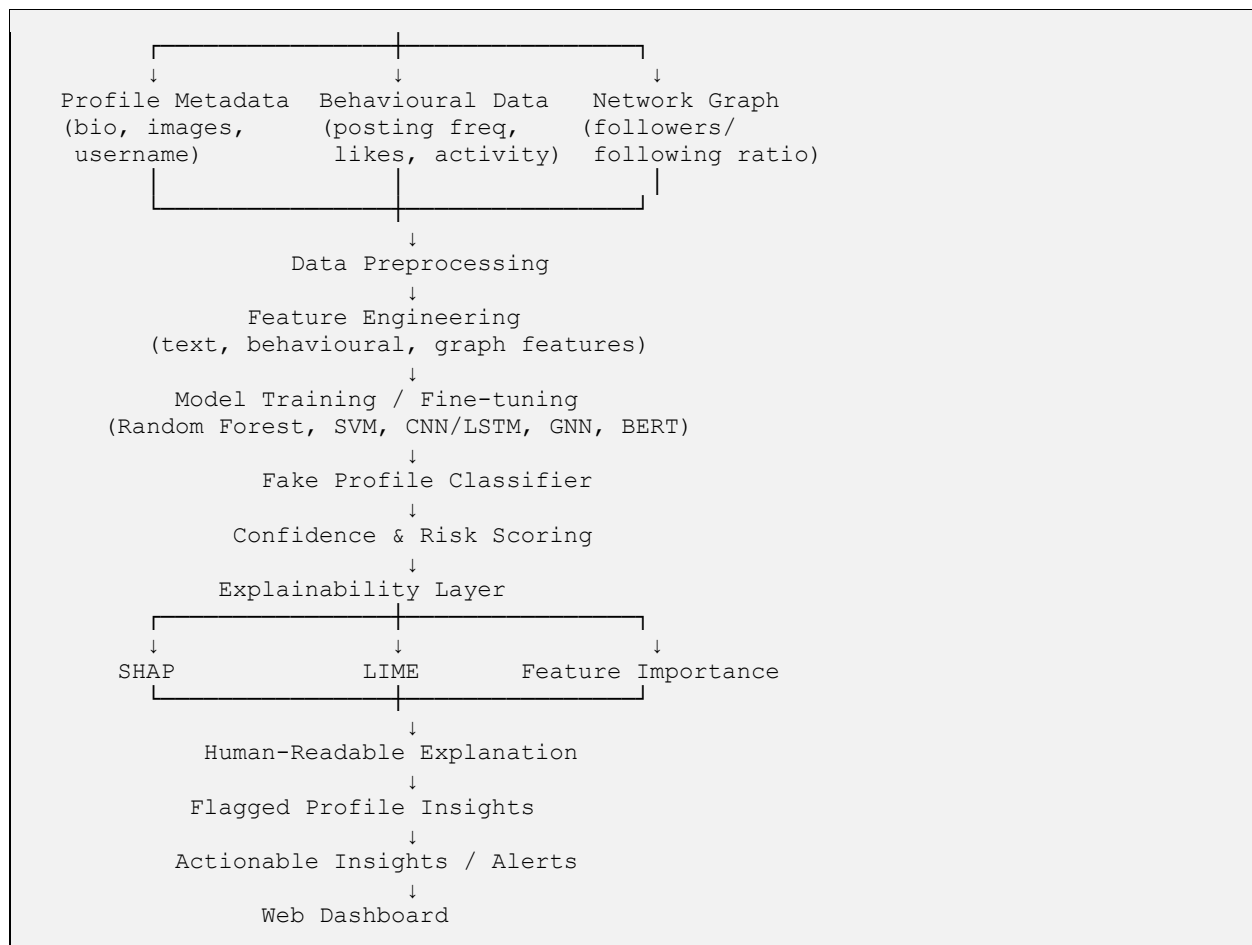
The system performs data preprocessing, feature extraction, model training, and classification to flag profiles as genuine or fake/bot, along with a confidence score indicating detection certainty. To improve trust and transparency, an explainability layer using techniques such as SHAP, LIME, or feature-importance analysis is incorporated so that the specific attributes driving a prediction can be understood by platform moderators and end users. Model performance is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC, while explainability outputs are assessed for consistency and interpretability. The resulting system aims to provide accurate, scalable, and explainable fake profile detection that supports content moderation, platform security, and safer user experiences on social networks.

Datasets

Dataset	URL link
Fake and Genuine Profile Dataset (Instagram)	https://www.kaggle.com/datasets/free4ever1/instagram-fake-spammer-genuine-accounts
Fake Social Media Account Detection Dataset	https://www.kaggle.com/datasets/satishkumarprajapati/fake-social-media-account-detection-dataset
Genuine/Fake User Profile Dataset (Twitter)	https://www.kaggle.com/datasets/whoseaspects/genuinefake-user-profile-dataset

Approach

MULTIPLE DATA SOURCES



Project Group Members

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